

Yolov8s-worldv2

- **Dataset:** Railroad v3 with 800 training images, 45 validation images, 25 test images via <https://universe.roboflow.com/>
- **Hardware:** T4 GPU
- **Training:** 50 epochs / image size 640

Yolov8s-worldv2 results (post training/validation scores)

```
Ultralytics 8.4.18 🚀 Python-3.12.12 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
YOLOv8s-worldv2 summary (fused): 88 layers, 12,749,288 parameters, 0 gradients, 33.1 GFLOPs
```

Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%	2/2 2.4it/s 0.8s
all	45	207	0.909	0.936	0.947	0.657	
railroad-crossing-down	26	45	0.974	0.827	0.932	0.683	
railroad-crossings	10	17	0.826	0.941	0.922	0.581	
railroad-sign	41	50	0.952	1	0.978	0.745	
railroad-sign-light-off	14	29	0.809	0.966	0.92	0.593	
railroad-sign-light-on	32	66	0.984	0.947	0.981	0.681	

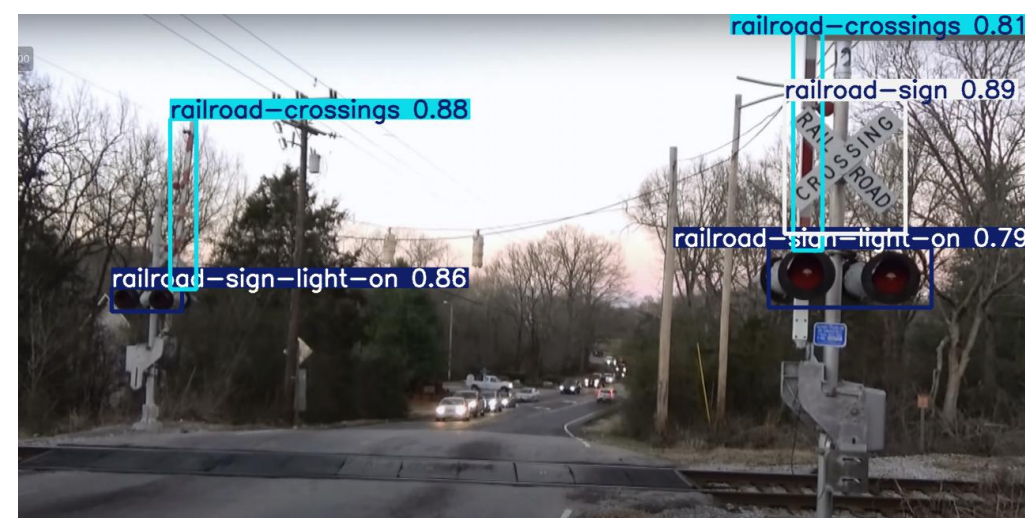
```
Speed: 0.2ms preprocess, 4.2ms inference, 0.0ms loss, 3.8ms postprocess per image
Results saved to /content/runs/detect/train
```

```
... Ultralytics 8.4.18 🚀 Python-3.12.12 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
val: Fast image access ✅ (ping: 0.4±0.1 ms, read: 72.9±22.7 MB/s, size: 165.6 KB)
val: Scanning /content/drive/MyDrive/Chuprov_Lab_Stuff/railroad_crossing_dataset/valid/labels.cache... 45 images, 1 backgrounds, 0 corrupt: 100% 45/45 11.1Mit/s 0.0s
```

Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%	3/3 1.0it/s 2.9s
all	45	207	0.909	0.936	0.945	0.657	
railroad-crossing-down	26	45	0.974	0.826	0.922	0.683	
railroad-crossings	10	17	0.826	0.941	0.922	0.584	
railroad-sign	41	50	0.952	1	0.978	0.744	
railroad-sign-light-off	14	29	0.809	0.966	0.92	0.589	
railroad-sign-light-on	32	66	0.984	0.947	0.981	0.683	

```
Speed: 6.5ms preprocess, 20.9ms inference, 0.0ms loss, 6.3ms postprocess per image
Results saved to /content/runs/detect/val
```

Yolov8s-worldv2 results (testing)



Yolov8m-worldv2 (training/validation scores)

- **Dataset:** Railroad v3 with 800 training images, 45 validation images, 25 test images via <https://universe.roboflow.com/>
- **Hardware:** T4 GPU
- **Training:** 50 epochs / image size 640

Yolov8m-worldv2 results (post training/validation)

```
Validating /content/runs/detect/train/weights/best.pt...
Ultralytics 8.4.18 🚀 Python-3.12.12 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
YOLOv8m-worldv2 summary (fused): 108 layers, 28,358,830 parameters, 0 gradients, 86.6 GFLOPs

```

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%	2/2 2.7it/s 0.7s
	all	45	207	0.927	0.94	0.943	0.675	
	railroad-crossing-down	26	45	0.942	0.822	0.921	0.731	
	railroad-crossings	10	17	0.873	0.941	0.907	0.616	
	railroad-sign	41	50	0.957	1	0.971	0.745	
	railroad-sign-light-off	14	29	0.878	0.99	0.94	0.586	
	railroad-sign-light-on	32	66	0.984	0.944	0.979	0.699	

```
Speed: 0.2ms preprocess, 8.7ms inference, 0.0ms loss, 1.8ms postprocess per image
Results saved to /content/runs/detect/train
```

```
Ultralytics 8.4.18 🚀 Python-3.12.12 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
val: Fast image access ✅ (ping: 0.4±0.1 ms, read: 60.6±16.6 MB/s, size: 165.6 KB)
val: Scanning /content/drive/MyDrive/Chuprova_Lab_Stuff/railroad_crossing_dataset/valid/labels.cache... 45 images, 1 backgrounds, 0 corrupt: 100% 45/45 17.2Mit/s 0.0s

```

	Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%	3/3 1.1it/s 2.8s
	all	45	207	0.927	0.94	0.943	0.671	
	railroad-crossing-down	26	45	0.942	0.822	0.921	0.728	
	railroad-crossings	10	17	0.873	0.941	0.907	0.601	
	railroad-sign	41	50	0.957	1	0.971	0.742	
	railroad-sign-light-off	14	29	0.878	0.99	0.94	0.585	
	railroad-sign-light-on	32	66	0.984	0.945	0.979	0.698	

```
Speed: 3.6ms preprocess, 33.9ms inference, 0.0ms loss, 3.3ms postprocess per image
Results saved to /content/runs/detect/val
```

Yolov8m-worldv2 results (testing)

